
CS 421 --- Regular Expression Activity

Name	Netid

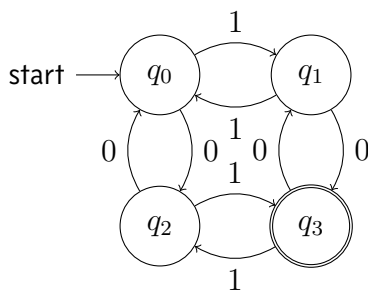
Please write your name/netid legibly in dark ink. Hand in one copy per team. Do not staple or mangle the corners.

Objectives

- Demonstrate the properties of regular languages.
- Demonstrate the correspondence between a Deterministic Finite Automata and a Right Linear Grammar.
- Identify languages that can be recognized by regular languages.
- Build an automata for a given regular language.

State Machines

Consider the following state machine:



Problem 1) Trace the following strings as inputs to the above state machine. Which strings are part of the language recognized by the state machine?

Input	Recognized?
0110	Y/N
00011	Y/N
0101	Y/N
11100010	Y/N

Problem 2) In English¹, describe the language this automata accepts.

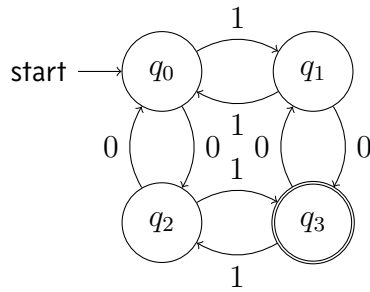
Problem 3) Is the language finite or infinite?

Problem 4) Is the amount of computation required to recognize or reject a string finite or infinite?

¹or what ever language you prefer

Correspondence to Right Linear Grammars

Here is that automata again, with an equivalent grammar.



$$\begin{aligned}
 S_0 &\rightarrow 1S_1|0S_2 \\
 S_1 &\rightarrow 1S_0|0S_3 \\
 S_2 &\rightarrow 1S_3|0S_0 \\
 S_3 &\rightarrow 1S_2|0S_1|\epsilon
 \end{aligned}$$

Problem 5) What do the S_n represent?

Problem 6) How is a transition modeled in the grammar?

Problem 7) The grammar is right linear, because there is at most one non-terminal symbol on the right hand side of any production. Suppose we added a rule like this one: $S_0 \rightarrow 1S_10S_2$. Could you still come up with a deterministic finite automata that matches the new grammar? Why or why not?

Categorization

Problem 8) Describe in English the following regular expressions

- `[a-zA-Z][a-zA-Z0-9]+`
- `[a-z]*(es|ed|ing)`
- `<[a-z0-9]+@[a-z0-9]+(\.[a-z0-9]+)+>`

Problem 9) Which of the following can be described by regular expressions?

- All the words in the English language
- All the Fibonacci numbers
- ``All Your Base Are Belong To Us'' video
- Numbers that are multiples of 4 (assume ≥ 2 digits)
- Words that have exactly as many as as they have bs
- Palindromes

Demo: Using grep and sed

Building an Automata

We can build an automata that recognizes integers that are multiples of 7!

Problem 10) To get started, fill out this table. The first two rows are done for you.

$n \bmod 7$	$10n \bmod 7$
0	0
1	3
2	
3	
4	
5	
6	

Problem 11) Now build your automata. If you are not sure how to get started, then ask yourself ``how many states will I need?" and ``what does a transition indicate?"